LINEAR ALGEBRA · LESSON 2

Elimination with Matrices

How every computer solves $Ax = b$: eliminate down to a triangle, then **back-substitute.**

Shown on a real money question: splitting ₹10 lakh across funds.

Pivots & multipliers | Portfolio mix | MIT 18.06, Lecture 2

THE PROBLEM

Pictures don't scale

In Lesson 1 we solved a portfolio by drawing lines and spotting tricks. With 50 funds and 50 rules, nobody can draw or guess.

The same three-fund system, now solved properly:

$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix} \begin{matrix} \text{budget} \\ \text{return} \\ \text{exposure} \end{matrix}$$

THE METHOD

Elimination: the way every software package solves equations.

THE IDEA

Triangles are easy

$$\begin{aligned} x + y + z &= 10 \\ -5y + 3z &= -9 \\ -0.4z &= -0.8 \end{aligned} \quad \Rightarrow \; z = 2,\; y = 3,\; x = 5$$

Solve the last equation, then work upwards. Elimination turns any good system into a triangle like this, **without changing its solution.**

TWO WORDS

Pivot: the diagonal entry used to clear the entries below it.
Multiplier: entry to clear ÷ pivot.

2 × 2

Two funds, one step

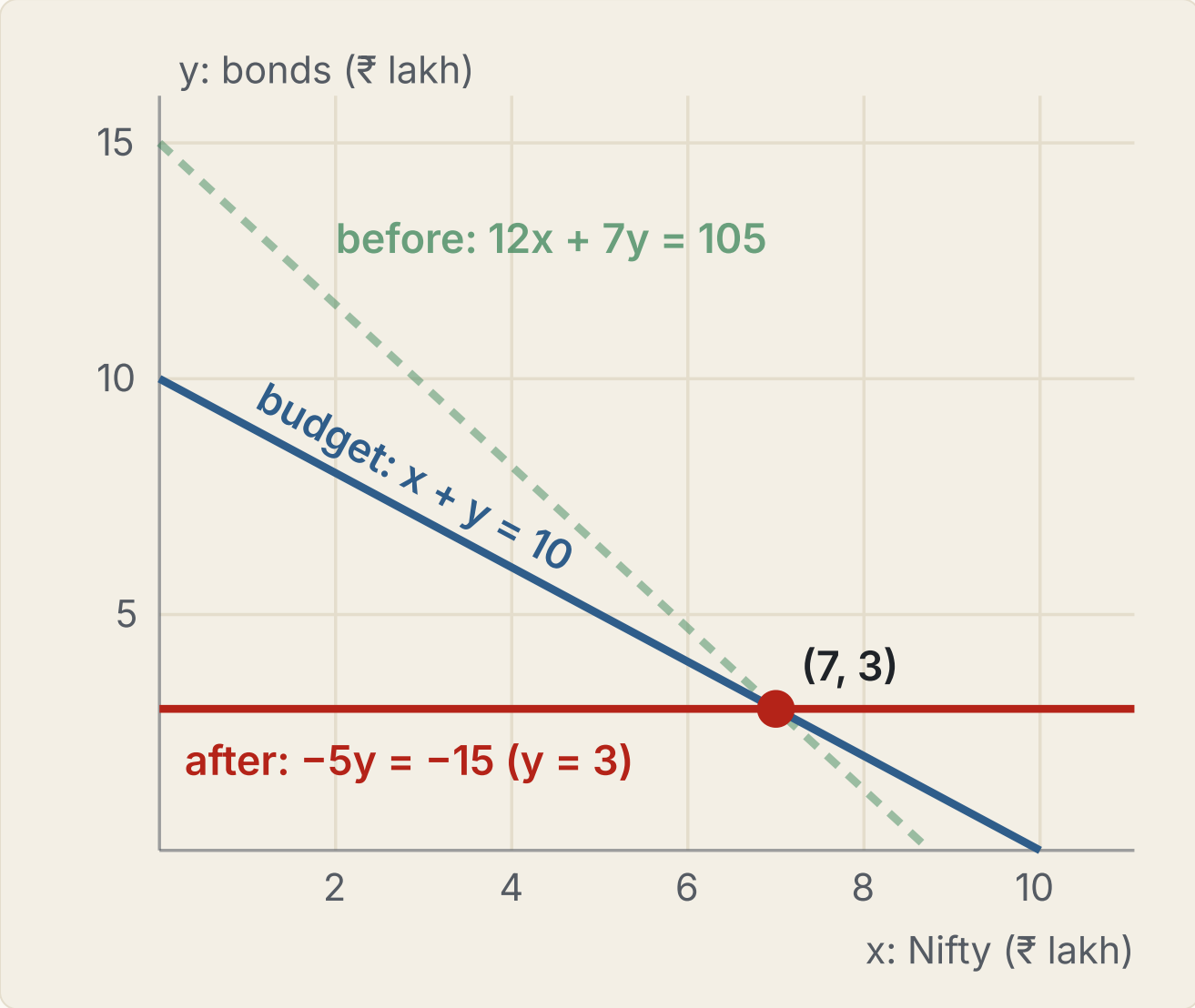
$$\left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 12 & 7 & 105 \end{array}\right] \xrightarrow{R_2 - 12R_1} \left[\begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 0 & \boxed{-5} & -15 \end{array}\right]$$

Multiplier $12/1 = 12$. Subtract $12 \times$ the budget rule from the return rule: $-5y = -15$.

So $y = 3$, then $x = 7$: **₹7 lakh Nifty, ₹3 lakh bonds.**

WHAT CHANGED?

New line, same answer



Elimination swaps the return line for $y = 3$. The crossing point $(7, 3)$ doesn't move.

Clear below the first pivot

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array}\right]$$

- $\ell_{21} = 12$: row 2 − 12 × row 1
- $\ell_{31} = 1$: row 3 − row 1

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & -5 & 3 & -9 \\ 0 & -1 & 0.2 & -2.6 \end{array}\right]$$

Clear below the second pivot

$\ell_{32} = (-1)/(-5) = 0.2$: row 3 − 0.2 × row 2.

$$[U \mid c] = \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & 3 & -9 \\ 0 & 0 & \boxed{-0.4} & -0.8 \end{array}\right]$$

RESULT

A became upper triangular U; b became c. Same solution. Pivots $1, -5, -0.4$ on the diagonal.

BACK SUBSTITUTION

Solve from the bottom up

1. $-0.4z = -0.8 \Rightarrow z = 2$
2. $-5y + 3(2) = -9 \Rightarrow y = 3$
3. $x + 3 + 2 = 10 \Rightarrow x = 5$

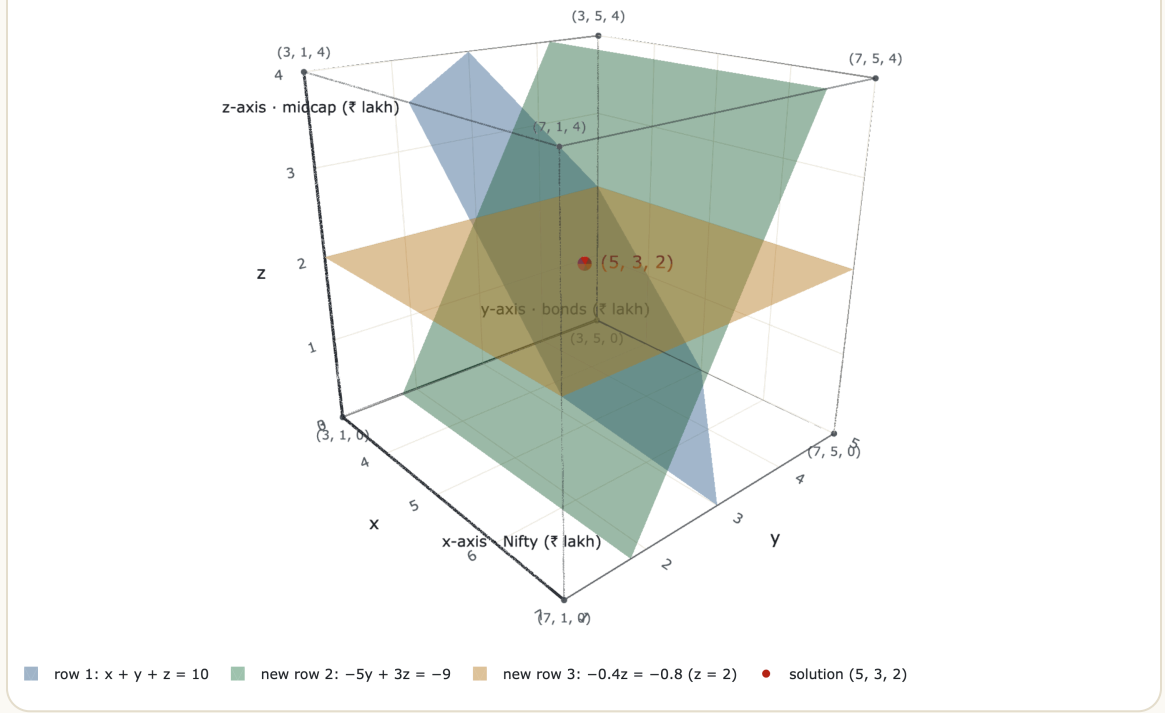
₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.

THE WHOLE METHOD

Forward elimination, then back substitution. That's all a computer does, with millions of rows.

THE ROW PICTURE

After elimination



New planes, ending with the flat $z = 2$. They still meet at the same point $(5, 3, 2)$.

WHEN IT BREAKS

A zero in the pivot position

Temporary: a non-zero entry below it? **Exchange rows** and carry on. (A credit fund earning 12% like Nifty causes this.)

Permanent: nothing non-zero below? No pivot: the matrix is **singular**. (A 50/50 balanced fund: the last row becomes $0 = -0.8$.)

PIVOTS DECIDE EVERYTHING

$n \times n$ matrix: n **pivots ⇔ invertible.** Fewer: no solution or infinitely many.

STEPS AS MATRICES

Each step is a multiplication

Row × matrix combines rows: $\begin{bmatrix} -12 & 1 & 0 \end{bmatrix} A =$ $-12\,(\text{row } 1) + (\text{row } 2)$.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad E_{32}E_{31}E_{21}A = U$$

Parentheses can move (associative), so $E = E_{32}E_{31}E_{21}$ does all of elimination at once.

BUT ORDER MATTERS

$AB \neq BA$ in general.

TWO MORE TOOLS

Swaps and undo

Permutation P: the identity with rows swapped. PA swaps rows of A; AP swaps columns.

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{adds back what } E_{21} \text{ subtracted}$$

WHY IT WORKS

Every step can be undone, so $Ax = b$ and $Ux = c$ have exactly the same solutions.

RECAP

Key takeaways

1. Elimination turns $Ax = b$ into triangular $Ux = c$, same solution.
2. Multiplier = entry ÷ pivot. Pivots can't be 0.
3. Back substitution: last row first.
4. Zero pivot: swap rows, or the matrix is singular.
5. Each step is a matrix: $E_{32}E_{31}E_{21}A = U$.
6. Steps are reversible, which is why it's right.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A lab that runs elimination step by step on your own numbers, 3D plots and worked examples:

learn.krisarj.com

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